DIAGNOSTIC WORKUP SUMMARY - DIFFERENTIAL ASSESSMENT

Synthetic patient: CASE-DIFF-8651 | Fictional name: Morgan Sample | Age: 67 years
Encounter: 2026-08-19 to 2026-08-21 | Fictional service: Acute Medical Assessment
Reason for review: six weeks of intermittent breathlessness and reduced exercise tolerance.
This is an invented test report. Differential entries below are attributed to the fictional clinical team and are not diagnoses or recommendations made by MedNLP Insight.

PRESENTING HISTORY

The patient described breathlessness when walking uphill and an intermittent dry cough. Symptoms were not reported at rest. No fever, purulent sputum, pleuritic pain, hemoptysis, orthopnea, or unilateral leg swelling was documented. The note records a 3 kg weight gain over two months, although a calibrated prior weight is unavailable. No recent surgery, immobility, or long-distance travel is recorded. Symptom severity is not quantified with a validated scale in the supplied documents.

BACKGROUND AND MEDICATION CONTEXT

History listed: hypertension, chronic obstructive pulmonary disease in an outside problem list, and remote tobacco exposure (approximately 22 pack-years; quit in 2016 per patient). No previous pulmonary-function test is attached. The patient reports taking tiotropium inhaler once daily and salbutamol inhaler as needed; adherence is not verified. No known history of venous thromboembolism is documented in the supplied record. Allergy section lists no known drug allergy, but reconciliation is marked incomplete.

INITIAL EXAMINATION AND OBSERVATIONS

Blood pressure 138/82 mmHg; pulse 88/min; respiratory rate 18/min; temperature 36.8 C; oxygen saturation 94% on room air. Mild expiratory wheeze was documented. No crackles, raised jugular venous pressure, or peripheral edema was recorded on that examination. Findings represent one assessment and do not establish the absence of intermittent signs.

DIFFERENTIAL ASSESSMENT - EVIDENCE REVIEW

The following possibilities are copied from the fictional team assessment. They remain differential considerations unless separately listed as a confirmed diagnosis.

1. Obstructive-airway symptom recurrence / COPD or asthma-related airflow limitation
Evidence recorded in support: intermittent wheeze on examination and prior obstructive airway disease in an outside problem list. The patient uses maintenance inhaler therapy.
Evidence recorded as limiting: no spirometry is available; baseline symptoms and inhaler adherence were not verified; no acute respiratory distress was observed in this assessment.
Status in note: possible contributor; not objectively confirmed during this encounter.

2. Cardiac dysfunction or volume-related cause of exertional breathlessness
Evidence recorded in support: patient-reported weight increase over two months and reduced exercise tolerance. NT-proBNP is 510 pg/mL; the laboratory report supplies no interpretive cutoff for this patient's age or setting. No prior echocardiogram is attached.
Evidence recorded as limiting: no peripheral edema or raised jugular venous pressure was documented; chest radiograph did not report pulmonary edema or pleural effusion.
Status in note: remains possible; no confirmed heart-failure diagnosis is made here.

3. Lower respiratory infection, including atypical infection
Evidence recorded in support: cough is present and C-reactive protein is 12 mg/L (reported reference 0-5 mg/L). Evidence recorded as limiting: no fever or purulent sputum was reported; white blood cell count is 7.4 10^9/L (4.0-11.0); chest radiograph shows no focal air-space consolidation. Viral panel was negative for the listed targets only.
Status in note: less supported by currently available findings; infection is not excluded.

DIFFERENTIAL ASSESSMENT - TESTS AND PLAN

4. Pulmonary vascular cause, including pulmonary embolism
Evidence recorded in support: exertional breathlessness is present. No D-dimer or CT pulmonary angiogram was performed in the supplied encounter. Evidence recorded as limiting: no pleuritic pain, hemoptysis, unilateral leg swelling, recent surgery, or known prior venous thromboembolism is documented. Status in note: not established; this record does not document a completed rule-out evaluation or confirmatory test.

5. Anemia or metabolic contributor
Hemoglobin 12.9 g/dL (laboratory reference 13.0-17.0), flagged low by the reporting lab. Sodium 139 mmol/L (135-145); creatinine 1.12 mg/dL (0.70-1.30); glucose 108 mg/dL (70-110). The clinician note lists anemia as a possible contributor, but states that this single mildly low measurement does not explain the full symptom history by itself. No iron profile, reticulocyte count, or prior hemoglobin trend is attached.

IMAGING AND OTHER RECORDED RESULTS

Chest radiograph: mild hyperinflation; no focal consolidation, pulmonary edema, or pleural effusion reported. ECG: sinus rhythm, no acute ischemic change described. Troponin below the reporting threshold. The report does not include echocardiography or pulmonary function testing. A negative finding on one test does not resolve every differential possibility.

DOCUMENTED PLAN AND OUTCOME

The team documented outpatient pulmonary-function testing and echocardiography as requested next steps. No appointment date or completed test result is present. The patient was discharged in stable condition with existing inhalers listed for continuation and primary-care review requested within 10 days. This summary does not state which differential diagnosis is correct. Do not infer a definitive cause from this fictional report; preserve the clinician's uncertainty.

END OF FICTIONAL DIFFERENTIAL WORKUP - NOT A REAL PATIENT RECORD OR MEDICAL ADVICE